

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Biogeochemical cycles: carbon, nitrogen, phosphorus - Set 1

Mark scheme

AQA spec 3.2.4; The physical environment; textbook pp. 148-157

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the correct answer. • 7×10^{11} tonnes (1)	1
01.2	A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown. • 10% of 10 000 = 1000 billion tonnes (1) • $1000 / 700 \times 100 = 143\%$ (accept 142.9) (1)	2
01.3	1 mark for each point. • Carbon is transferred into and out of rocks extremely slowly (fossilisation / sedimentation, weathering, volcanic release) (1) • Residence time is millions of years so the flux to the atmosphere is tiny compared with respiration / photosynthesis / combustion (1)	2
01.4	1 mark for each point. • CO ₂ dissolves in the sea (forming carbonic acid / hydrogen carbonate ions) and is absorbed by phytoplankton through photosynthesis (1) • Organisms such as molluscs, corals and planktonic organisms build calcium carbonate shells / exoskeletons (1) • When they die the shells sink and sediment on the seabed, becoming limestone / chalk over long periods (1)	3
01.5	1 mark for each point. • Aerobic respiration uses oxygen to break down organic compounds completely to CO ₂ (and water), releasing much more energy (1) • Without oxygen, anaerobic organisms only partly break down organic compounds, releasing carbon as methane (CH ₄), which is later oxidised to CO ₂ in the atmosphere (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-149; AQA spec 3.2.4.2.	

Question	Marking guidance	Marks
02.1	2 marks for each point (partial credit for a partly correct point). • Ploughing: increases the oxygen supply to soil decomposers so aerobic decomposition of dead organic matter is faster; carbon in the soil DOM store falls and atmospheric CO₂ rises (2) • Draining: waterlogged anaerobic conditions preserved dead organic matter (peat) and produced methane slowly; drainage makes the soil aerobic so stored carbon is rapidly decomposed to CO₂ (though methane release stops) (2) • Marine pollution: kills phytoplankton so less dissolved CO₂ is absorbed by photosynthesis and less carbon sinks to sediments in dead plankton; less CO₂ then dissolves from the atmosphere to replace it, so atmospheric CO₂ rises (2)	6
02.2	Any 3 from the points below - 1 mark each, maximum 3. • Rice paddy fields (1) • Landfill sites (1) • Anaerobic sediments in reservoirs (1) • Livestock intestines (cattle) (1) • Manure / slurry stores (1)	3
02.3	1 mark. • More CO₂ dissolves in the sea, forming carbonic acid which dissociates into hydrogen carbonate and hydrogen ions; more H⁺ lowers the pH (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 150-151; AQA spec 3.2.4.2.	
03.1	1 mark for each point. • Trees absorb CO₂ by photosynthesis and store carbon in wood as they grow (1) • Once mature there is no further net storage (photosynthesis balanced by respiration / decomposition), though the standing forest remains a reservoir (1) • Harvesting the timber and keeping it as wood (eg house construction) then replanting produces additional storage as the new trees grow (1)	3
03.2	1 mark for each point. • Pre-combustion: the fuel is modified before use so no CO₂ is released when burnt - eg gasification of coal to hydrogen and CO₂ (CO₂ stored, hydrogen distributed); oxy-fuel combustion in pure oxygen giving only CO₂ and water vapour which is condensed out (1) • Advantage: capture at a large central facility; suits small dispersed sources (vehicles) that cannot capture CO₂ (1) • Post-combustion: CO₂ removed from exhaust gases by solvents, membrane filtration, adsorption / desorption, cryogenic separation, graphene (1) • Disadvantage: only about 20% of flue gas is CO₂ so separation is costly / energy-intensive; none fully developed (1)	4
03.3	Any 3 from the points below - 1 mark each, maximum 3. • Injected into depleted oil / gas fields or saline aquifers (1) • Injected into oil reservoirs to provide pressure for secondary oil recovery (1) • Reaction with fractured basalt to form solid carbonate minerals (1) • Concern: leakage back to the atmosphere / monitoring over centuries; limited suitable geology; cost (1)	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 151-152; AQA spec 3.2.4.2.	
04.1	<i>1 mark for each point.</i> • 1 nitrogen fixation; 2 nitrification (1) • 3 denitrification; 4 ammonification (1) • 5 ionisation; 6 leaching (3 marks for all six, 2 for four or five, 1 for two or three) (1)	3
04.2	<i>1 mark for each point.</i> • Drainage makes the soil aerobic (1) • Aerobic nitrifying bacteria (and nitrogen fixers) increase so more ammonium is converted to nitrate (1) • Anaerobic denitrifying bacteria decline so less nitrate is lost as nitrogen gas (1)	3
04.3	<i>1 mark for each point.</i> • Nitrogen gas is very unreactive so most organisms cannot use it directly (1) • Only nitrogen fixation (bacteria, lightning, Haber process) converts it to usable ammonium / nitrate; the cycle recycles a small usable amount (1)	2
04.4	<i>1 mark for each point.</i> • Rhizobium bacteria in their root nodules fix atmospheric nitrogen into ammonium / proteins (1) • When the plants die / are ploughed in, decomposition and nitrification release nitrates for the following crop (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 152-154; AQA spec 3.2.4.3.	
05.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Human = $120 + 60 + 30 = 210$; total = 325 (1) • $210 / 325 \times 100 = 64.6\%$ (accept 65%) (1)	2
05.2	<i>1 mark for each point.</i> • Nitrogen and hydrogen react to form ammonia over an iron catalyst at high temperature and pressure; ammonia then converted to nitrates for fertiliser (1) • Uses large amounts of energy from fossil fuels (also the hydrogen source), releasing CO₂ and NO_x (1) • Biological fixation by bacteria occurs at low temperature using solar energy captured by plants (enzymes lower activation energy) (1)	3
05.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Eutrophication of water bodies by leached nitrates / sewage (1) • Photochemical smogs and tropospheric ozone from NO_x (1) • Acid rain (nitric acid) (1) • Nitrous oxide as a greenhouse gas (1) • NO_x toxicity / respiratory disease (1) • Nitrate in drinking water (methaemoglobinaemia) (1)	3
05.4	<i>1 mark for each point.</i> • Carbon is lost as CO₂ (composting) or CH₄ (digestion) so the C:N ratio improves / more nitrogen-rich (1) • Bulk and odour reduced, pathogens killed, easier to apply; digestion also yields biogas (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 154-155; AQA spec 3.2.4.3.	
06.1	<i>1 mark for each point.</i> • There is no gaseous / atmospheric reservoir of phosphorus (no equivalent of nitrogen fixation) (1) • Most phosphorus compounds have low solubility so little is available in soil water (1) • Phosphorus lost to marine sediments only returns through very slow processes: mountain building (uplift) and weathering of rocks (1)	3
06.2	<i>1 mark for each point.</i> • Fungal hyphae form a symbiotic relationship with plant roots and spread through a large volume of soil (1) • Greatly increasing the surface area for absorbing (poorly mobile) phosphate ions, passed to the plant in exchange for carbohydrates (1)	2
06.3	<i>1 mark for each point.</i> • Mining of calcium phosphate rock (and past mining of guano deposits, eg Peru, Nauru, now largely exhausted) (1) • Treated to make more soluble ammonium phosphate fertilisers, mobilising phosphorus into the environment (1) • Fertiliser runoff / eroded soil / sewage effluent carry phosphates into water bodies causing eutrophication (with nitrates); harvesting removes phosphorus from farmland (1)	3
06.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Use biological wastes (manure, sewage sludge, compost) as fertiliser to return phosphates rather than mining rock (1) • Breed crops that absorb phosphate more efficiently (1) • Maintain conditions for mycorrhizal fungi (less tillage, fewer fungicides, organic matter) (1) • Recover phosphate from sewage effluent by iron(III) sulfate precipitation; reduce soil erosion (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 156-157; AQA spec 3.2.4.4.	

Question	Marking guidance	Marks
07.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Carbon: combustion of fossil fuels and biomass moving carbon from long-term stores to the atmosphere; deforestation reducing photosynthesis; ploughing / drainage increasing aerobic decomposition; new anaerobic environments (paddies, landfill, reservoirs, livestock) releasing methane; marine pollution reducing phytoplankton; consequences: rising CO₂ and CH₄, climate change, ocean acidification, reduced soil organic matter • Nitrogen: Haber process doubling fixation; legume cultivation; drainage increasing nitrification and reducing denitrification; combustion NO_x; sewage and fertiliser leaching; consequences: eutrophication, smogs, acid rain, N₂O warming, NO_x toxicity • Counter-strategies carbon: renewables / nuclear replacing fossil fuels, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, increasing soil organic matter (low tillage, mulching), CCS (pre / post combustion, storage) • Counter-strategies nitrogen: reduced combustion, catalytic converters / urea sprays, biological fixation instead of Haber, legumes and crop rotation, management of biological wastes as fertiliser, low-solubility fertilisers, timing, buffer strips, low tillage, nitrate vulnerable zones • Evaluation: some strategies well established and cheap (rotation, buffer strips), others expensive or unproven (CCS); scale of fossil fuel dependence; feedback and time lags; link to circular economy and low-temperature natural processes; judgement	9
	Model answer - Level 3 (8 marks)	
	Humans have shifted the carbon cycle by burning fossil fuels and biomass, moving carbon from long-term stores into the atmosphere faster than photosynthesis and ocean uptake remove it, while deforestation reduces photosynthesis, ploughing and drainage speed aerobic decomposition, and paddies, landfills and livestock create new anaerobic sources of methane. The nitrogen cycle has changed as much: the Haber process has roughly doubled fixation, legume crops add more, drainage increases nitrification and reduces denitrification, combustion produces NO_x, and leached fertiliser and sewage nitrate cause eutrophication, while nitrous oxide adds to warming.	

Question	Marking guidance	Marks
	Carbon can be counteracted by replacing fossil fuels with renewables and nuclear power, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, and increasing soil organic matter through low tillage and mulching, with carbon capture and storage as an unproven option. For nitrogen, catalytic converters and urea sprays reduce combustion NO _x , biological fixation by legumes and rotation can replace manufactured nitrate, biological wastes can be managed as fertiliser, and low-solubility fertilisers, careful timing and buffer strips reduce leaching.	
	Some strategies are cheap and established - rotation, buffer strips, fertiliser timing - but others are expensive or untested, and the scale of fossil-fuel dependence, together with feedback and time lags, means the atmosphere will respond slowly. The strategies that mimic natural low-temperature cycling, as in the circular economy, are the most sustainable; the most effective is simply to leave carbon in the ground.	
	Weaker answer - Level 1 (3 marks)	
	Humans have changed the carbon cycle by burning fossil fuels which releases carbon dioxide, and by cutting down trees. They have changed the nitrogen cycle by using fertilisers which run off into rivers and cause eutrophication. To counteract this we could use renewable energy instead of fossil fuels and plant more trees. We could use less fertiliser. These would help the cycles go back to normal.	
	What makes the difference	
	• Level 1 names one or two changes per cycle; Level 3 covers several processes for each (decomposition, methane sources, Haber process, nitrification, NO _x).	
	• Level 3 connects each counter-strategy to the process it reverses; Level 1's strategies are general.	
	• The evaluation in Level 3 distinguishes established from unproven strategies and comments on scale and time lags - Level 1 has no evaluation.	
07.2	<i>1 mark.</i> • The average time an atom / molecule remains in the reservoir before being transferred out (volume / transfer rate) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-157; AQA spec 3.2.4.	
08.1	<i>1 mark for the correct answer.</i> • $(120 + 95) - (2 + 45) = 168 \text{ t ha}^{-1}$ (1)	1
08.2	<i>1 mark for each point.</i> • Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO ₂ (1) • Harvest removes biomass so less dead organic matter is returned; pasture roots and litter constantly add organic matter (1)	2
08.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Low / zero tillage (1) • Adding manure, compost or mulch (1) • Growing cover crops / leaving crop residues (1) • Converting to long-term grass or tree crops in rotation (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	